

Class 10: Structural Bioinformatics (pt1)

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Introduction to the RCSB Protein Data Bank (PDB)

PDB statistics

```
pdb <- read.csv("Data Export Summary.csv")
pdb
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	180,758	23,111	12,813	348	229
2	Protein/Oligosaccharide	10,488	3,741	34	8	11
3	Protein/NA	9,205	6,751	287	26	8
4	Nucleic acid (only)	3,154	250	1,578	3	15
5	Other	178	27	35	4	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron	Other	Total			
1	84	32	217,375			
2	1	0	14,283			
3	0	0	16,277			
4	3	1	5,004			
5	0	0	244			
6	0	4	22			

```
pdb$X.ray
```

```
[1] "180,758" "10,488" "9,205" "3,154" "178" "11"
```

They print out above `pdb$X.ray` is “character” not “numeric”. Therefore I can’t do math with it. We need to fix this..

Two functions that can help here are `sub()` and `as.numeric()`.

```
#We want to get rid of (or sub out) commas:  
as.numeric(sub(",", "", x=pdb$X.ray))
```

```
[1] 180758 10488 9205 3154 178 11
```

We could make a function to do this:

```
rm.comma <- function(x) {  
  tmp <- sub(",", "", x)  
  sum (as.numeric(tmp))  
}  
rm.comma(pdb$X.ray)
```

```
[1] 203794
```

We could also use a different import function for this CSV that speaks American. (i.e. deals with commas in numbers in a comma separated value file)

```
library(readr)  
pdb_stats <- read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Molecular Type
```

```
dbl (4): Integrative, Multiple methods, Neutron, Other
```

```
num (4): X-ray, EM, NMR, Total
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

Counting **all the structures (not just proteins)**, we have to add up all the numbers for each column.

```
total_num <- sum(pdb_stats$Total)
X_ray_num <- sum(pdb_stats$`X-ray`)
EM_num <- sum(pdb_stats$EM)
cat("X-ray percentage: ", 100*X_ray_num/total_num, "%\n", sep="")
```

X-ray percentage: 80.48577%

```
cat("EM percentage: ", 100*EM_num/total_num, "%", sep="")
```

EM percentage: 13.38046%

Q2: What proportion of structures in the PDB are protein?

```
Protein_only_num <- pdb_stats$Total[1]
cat("Protein-only percentage: ", 100*Protein_only_num/total_num, "%\n", sep="")
```

Protein-only percentage: 85.84941%

```
Protein_total_num <- sum(pdb_stats$Total[1:3])
cat("Protein-total percentage: ", 100*Protein_total_num/total_num, "%", sep="")
```

Protein-total percentage: 97.91868%

The total number of protein sequences is 202,556,314 from UniProt statistics.

```
100*Protein_only_num/202556314
```

[1] 0.1073158

Key-point: We have a very, very small structural coverage of known proteins (~0.1%). Most structures we know about (~80%) come from one method (X-ray crystallography)

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB? (Skip)

Professor had skipped this Q, but still the search results gave me the number of 1,227 structures.

Visualizing PDB data with Mol-star

Main stand alone web cersion with all features is at <https://molstar.org/viewer/>.

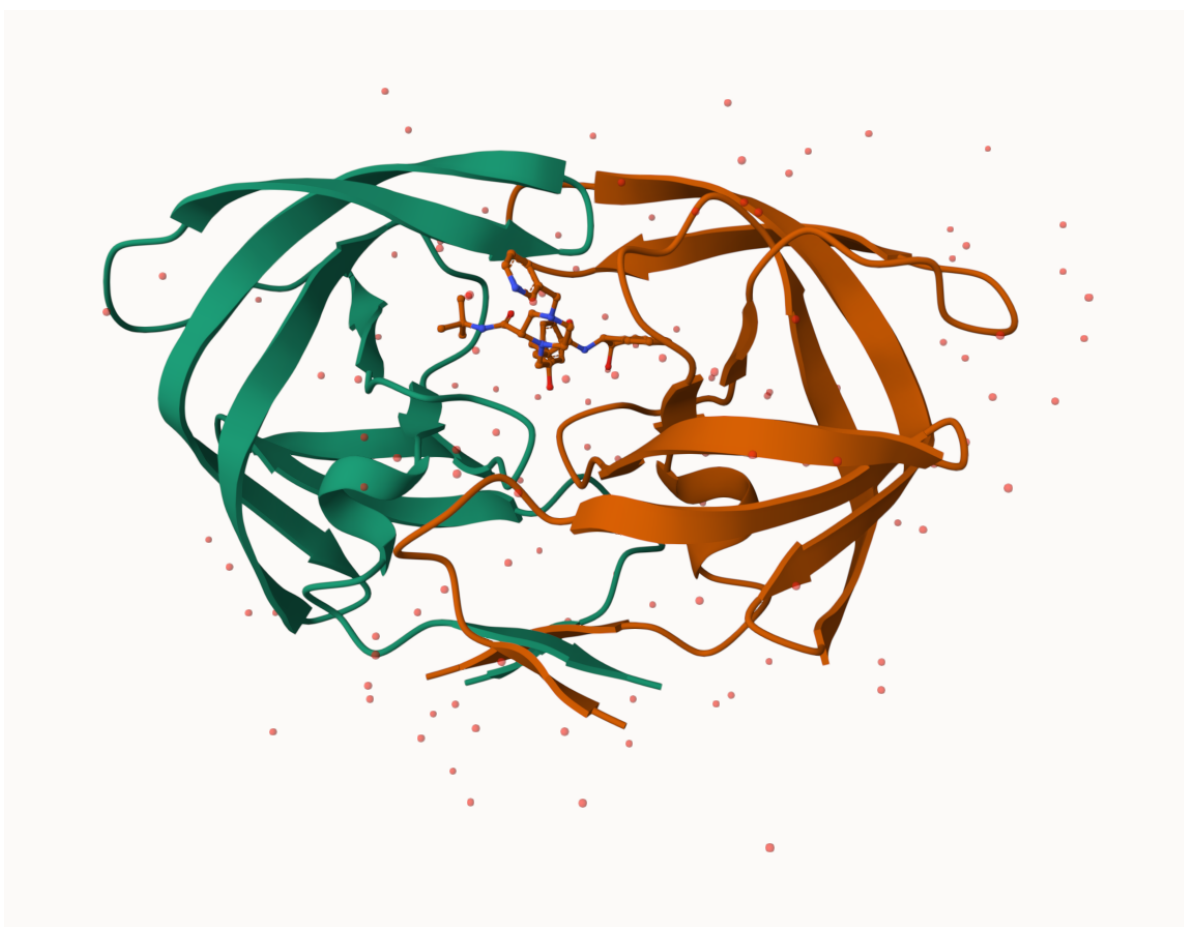
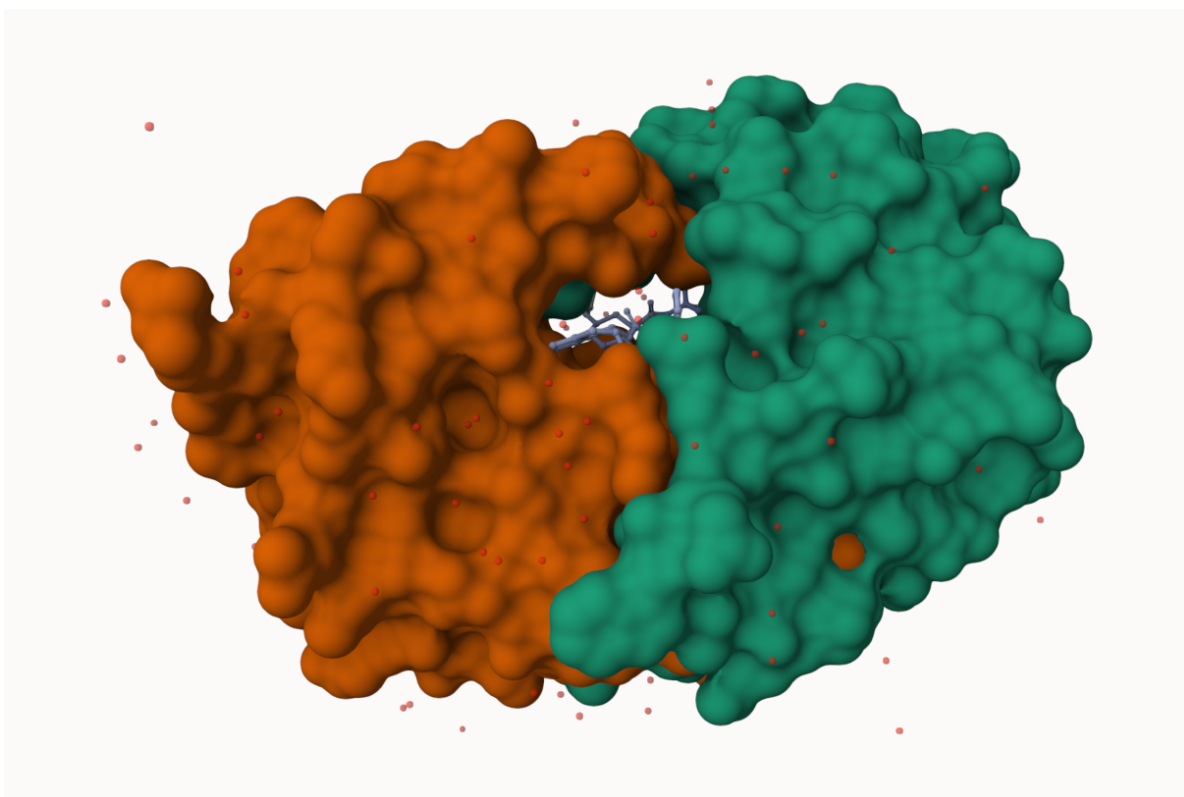


Figure 1: The HIV-protease enzyme is a homodimer with two chains



> Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Only the Oxygen atom is indicated. (hydrogen atoms are not depicted)

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

308. (“HOH308”)

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

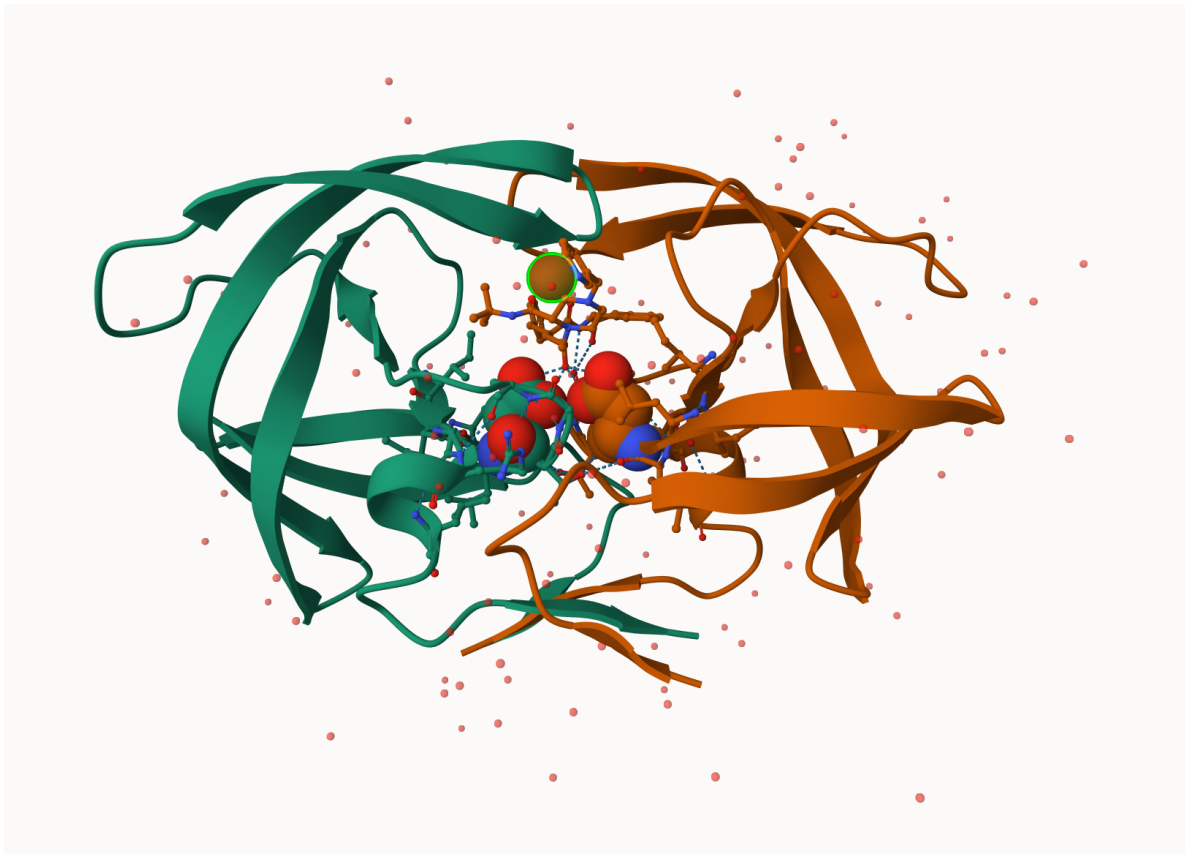


Figure 2: Spacefill display of catalytic ASP 25 amino acids and key binding site water molecule

Getting started with the Bio3D package

Bio3D is a R package from CRAN for structural bioinformatics

```
library(bio3d)
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```

Total Models#: 1
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

```

Protein sequence:

```

PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF

```

```

+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call

```

Q7: How many amino acid residues are there in this pdb object?

There are 198 amino acid residues (as we can see from the Calpha atoms count).

Q8: Name one of the two non-protein residues?

HOH = water.

Q9: How many protein chains are in this structure?

2 chains. (A and B)

```
attributes(pdb)
```

```
$names
```

```
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

```
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62

```

3 ATOM      3      C <NA>  PRO      A      1      <NA> 29.760 38.071 4.022 1 42.64
4 ATOM      4      O <NA>  PRO      A      1      <NA> 28.600 38.302 3.676 1 43.40
5 ATOM      5     CB <NA>  PRO      A      1      <NA> 30.508 37.541 6.342 1 37.87
6 ATOM      6     CG <NA>  PRO      A      1      <NA> 29.296 37.591 7.162 1 38.40
      segid elesy charge
1  <NA>      N  <NA>
2  <NA>      C  <NA>
3  <NA>      C  <NA>
4  <NA>      O  <NA>
5  <NA>      C  <NA>
6  <NA>      C  <NA>

```

There are lots of functions that we can work with these `pdb` objects

```
head(pdbseq(pdb))
```

```

      1      2      3      4      5      6
"P" "Q" "I" "T" "L" "W"

```

We can have a quick interactive view of any of these :

```
library(bio3dview)
#view.pdb(pdb)
```

Let's try a custom view

```
#view.pdb(pdb,colorScheme="sse", backgroundColor="black")
```

Q. Create a custom view of HIV-Pr highlighting the active site ASP (`resno=25`), the two chains (in your choice of colors) and the ligand all on a custom color background?

```

library(NGLViewer)
active.site <- atom.select(pdb, resno=25)
#view.pdb(pdb,
#         cols=c("red", "blue"),
#         highlight=active.site,
#         highlight.style="spacefill",
#         backgroundColor="pink") |>
# setRock()

```


Predict the flexibility of a given structure

Let's do a Normal Mode Analysis (NMA) to predict the flexibility of a given `pdb` object:

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, `rm.alt=TRUE`

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

Protein sequence:

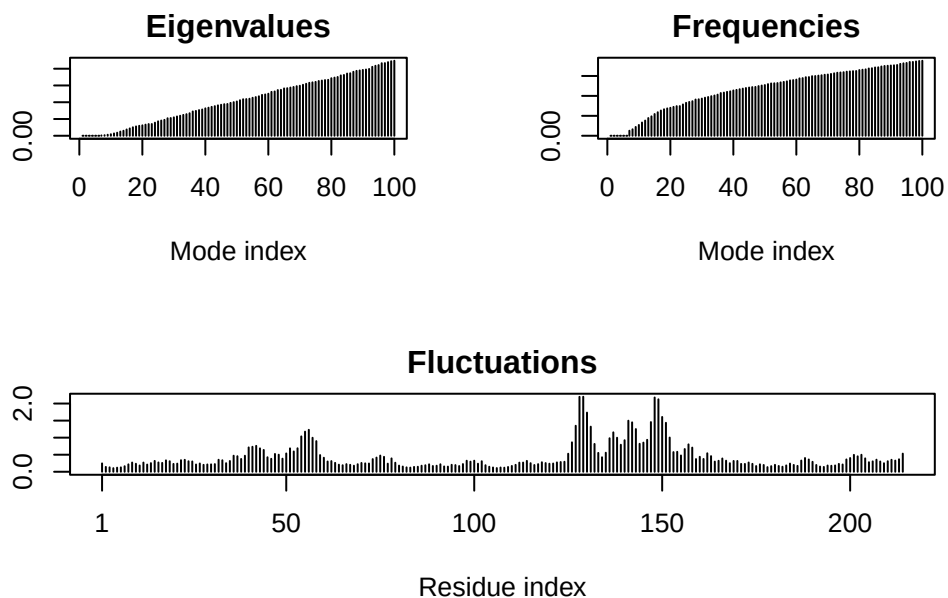
```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV
TDELVIALVKERIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

```
m <- nma(adk)
```

```
Building Hessian...      Done in 0.012 seconds.
Diagonalizing Hessian... Done in 0.276 seconds.
```

```
plot(m)
```



```
#view.nma(m)
```

Write out the results for viewing in Mol-star:

```
mktrj(m, file="nma.pdb")
```

Comparative structure analysis of the ADK family

Q10. Which of the packages above is found only on BioConductor and not CRAN?

msa

Q11. Which of the above packages is not found on BioConductor or CRAN?

bio3dview

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True

Our first step is to find a sequence for this family. We will use the database ID “1ake_A” here:

```
id <- "1ake_A"
aa <-get.seq(id)
```

Warning in get.seq(id): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
aa
```

```

      1      .      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      .      60

      61      .      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRTPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      .      120

      121      .      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTRKDDQEETVRKRLVEYHQMTAPLIG
      121      .      .      .      .      .      .      180

      181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
      181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

214 amino acids.

Search for related sequences in the database

```
blast <- blast.pdb(aa)
```

Searching ... please wait (updates every 5 seconds) RID = 02GMBRJ5016

.....

Reporting 96 hits

```
head(blast$hit.tbl)
```

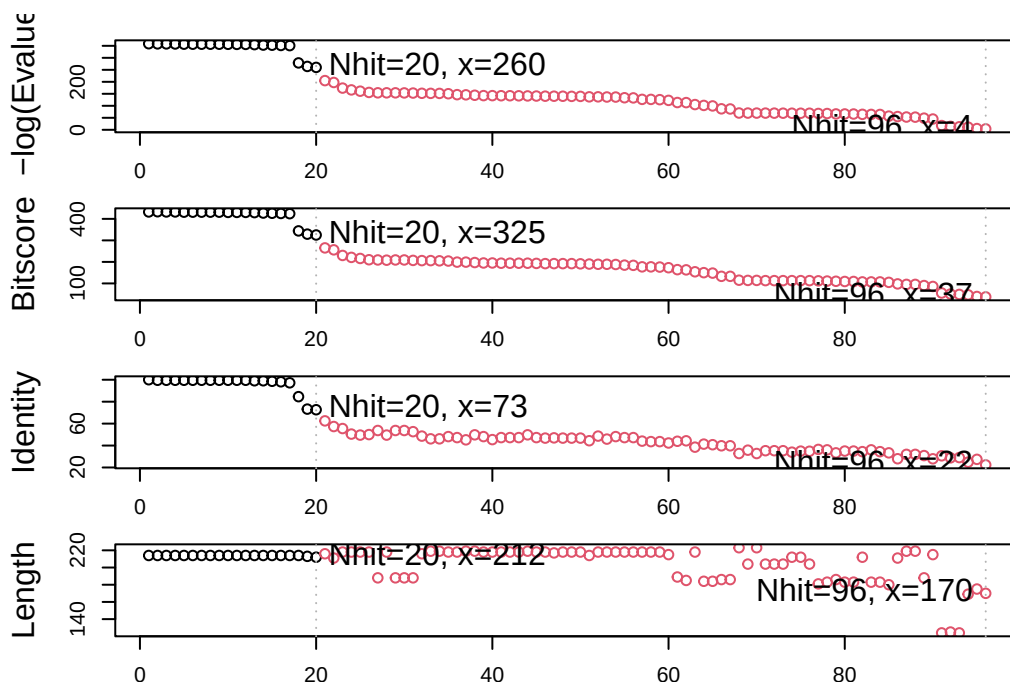
	queryid	subjectids	identity	alignmentlength	mismatches	gapopens	q.start
1	Query_7064075	1AKE_A	100.000	214	0	0	1
2	Query_7064075	8BQF_A	99.533	214	1	0	1
3	Query_7064075	4X8M_A	99.533	214	1	0	1
4	Query_7064075	6S36_A	99.533	214	1	0	1
5	Query_7064075	9R6U_A	99.533	214	1	0	1
6	Query_7064075	9R71_A	99.533	214	1	0	1

	q.end	s.start	s.end	evaluate	bitscore	positives	mlog.evaluate	pdb.id	acc
1	214	1	214	1.82e-156	432	100.00	358.6044	1AKE_A	1AKE_A
2	214	21	234	2.97e-156	433	100.00	358.1147	8BQF_A	8BQF_A
3	214	1	214	3.25e-156	432	100.00	358.0246	4X8M_A	4X8M_A
4	214	1	214	4.77e-156	432	100.00	357.6409	6S36_A	6S36_A
5	214	1	214	1.06e-155	431	99.53	356.8424	9R6U_A	9R6U_A
6	214	1	214	1.25e-155	431	99.53	356.6775	9R71_A	9R71_A

```
hits <- plot(blast)
```

```
* Possible cutoff values: 260 3
    Yielding Nhits:      20 96
```

```
* Chosen cutoff value of: 260
    Yielding Nhits:      20
```



```
hits$pdb.id
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A" "8Q2B_A" "8RJ9_A"
[9] "6RZE_A" "4X8H_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A" "6HAP_A"
[17] "6HAM_A" "8PVW_A" "4K46_A" "4NP6_A"
```

```
files <- get.pdb(hits$pdb.id, path="pdb", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/8BQF.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/4X8M.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6S36.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R6U.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R71.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8Q2B.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8RJ9.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8H.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

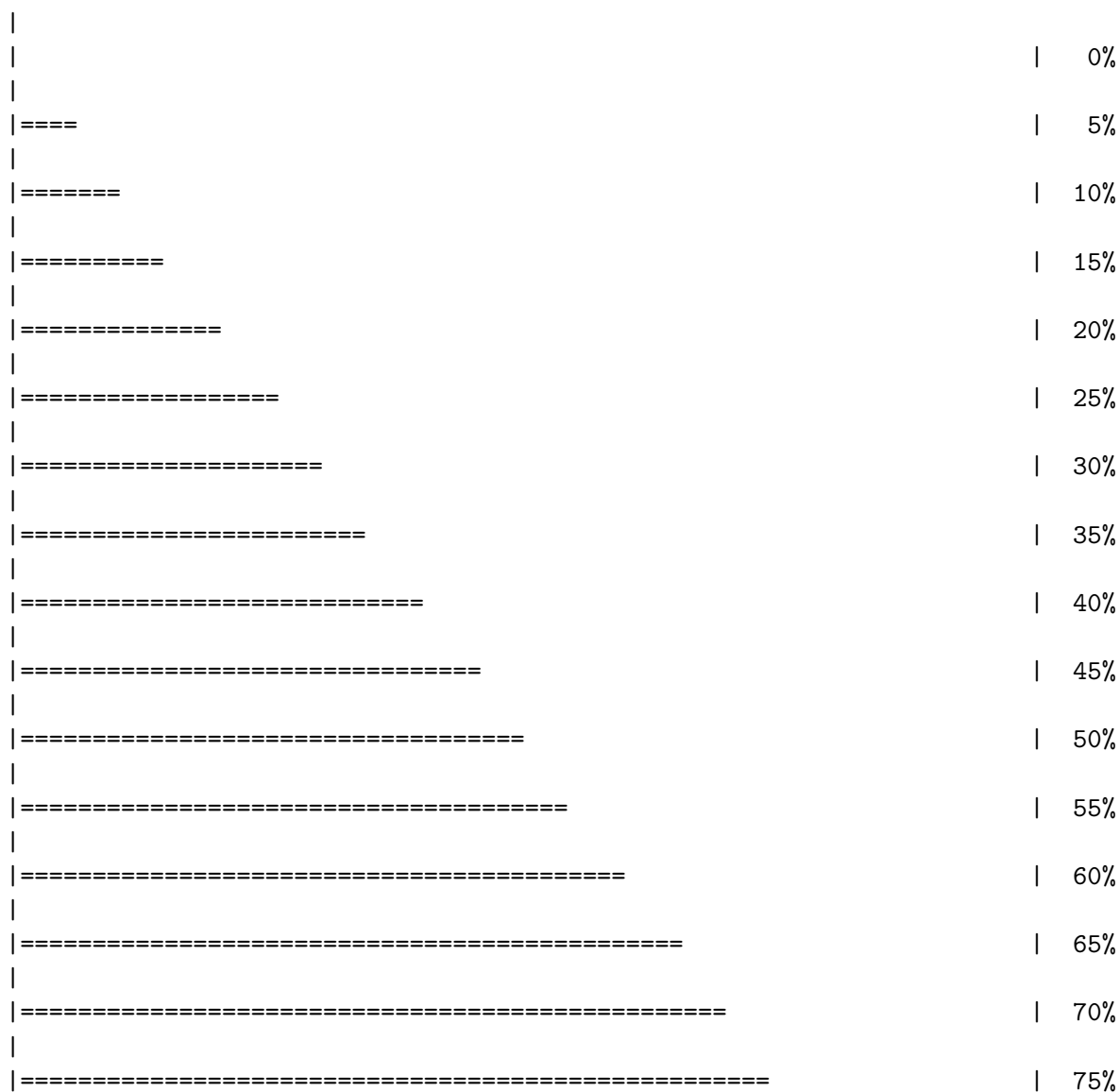
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8PVW.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4NP6.pdb.gz exists. Skipping download



```

|
|=====| 80%
|
|=====| 85%
|
|=====| 90%
|
|=====| 95%
|
|=====| 100%

```

Align and superpose all these ADK structures

```
pdbbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```

pdbbs/split_chain/1AKE_A.pdb
pdbbs/split_chain/8BQF_A.pdb
pdbbs/split_chain/4X8M_A.pdb
pdbbs/split_chain/6S36_A.pdb
pdbbs/split_chain/9R6U_A.pdb
pdbbs/split_chain/9R71_A.pdb
pdbbs/split_chain/8Q2B_A.pdb
pdbbs/split_chain/8RJ9_A.pdb
pdbbs/split_chain/6RZE_A.pdb
pdbbs/split_chain/4X8H_A.pdb
pdbbs/split_chain/3HPR_A.pdb
pdbbs/split_chain/1E4V_A.pdb
pdbbs/split_chain/5EJE_A.pdb
pdbbs/split_chain/1E4Y_A.pdb
pdbbs/split_chain/3X2S_A.pdb
pdbbs/split_chain/6HAP_A.pdb
pdbbs/split_chain/6HAM_A.pdb
pdbbs/split_chain/8PVW_A.pdb
pdbbs/split_chain/4K46_A.pdb
pdbbs/split_chain/4NP6_A.pdb

```

```

    PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE

```



```

.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..

```

Extracting sequences

```

pdb/seq: 1   name: pdbc/split_chain/1AKE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbc/split_chain/8BQF_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbc/split_chain/4X8M_A.pdb
pdb/seq: 4   name: pdbc/split_chain/6S36_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbc/split_chain/9R6U_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6   name: pdbc/split_chain/9R71_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbc/split_chain/8Q2B_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8   name: pdbc/split_chain/8RJ9_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9   name: pdbc/split_chain/6RZE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10  name: pdbc/split_chain/4X8H_A.pdb
pdb/seq: 11  name: pdbc/split_chain/3HPR_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbc/split_chain/1E4V_A.pdb
pdb/seq: 13  name: pdbc/split_chain/5EJE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14  name: pdbc/split_chain/1E4Y_A.pdb
pdb/seq: 15  name: pdbc/split_chain/3X2S_A.pdb
pdb/seq: 16  name: pdbc/split_chain/6HAP_A.pdb
pdb/seq: 17  name: pdbc/split_chain/6HAM_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18  name: pdbc/split_chain/8PVW_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 19  name: pdbc/split_chain/4K46_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE

```

pdb/seq: 20 name: pdb/split_chain/4NP6_A.pdb

pdb

	1	.	.	.	40
[Truncated_Name:1] 1AKE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:2] 8BQF_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:3] 4X8M_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:4] 6S36_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:5] 9R6U_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:6] 9R71_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:7] 8Q2B_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:8] 8RJ9_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:9] 6RZE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:10] 4X8H_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:11] 3HPR_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:12] 1E4V_A.pdb	--MRIILLGAPVAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:13] 5EJE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:14] 1E4Y_A.pdb	--MRIILLGALVAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:15] 3X2S_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:16] 6HAP_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:17] 6HAM_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:18] 8PVW_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:19] 4K46_A.pdb	--MRIILLGAPGAGKGTQAQFIMAKFGIPQISTG	DMLRAA			
[Truncated_Name:20] 4NP6_A.pdb	NAMRIILLGAPGAGKGTQAQFIMEKFGIPQISTG	DMLRAA			
	*****	*****	*~*****		
	1	.	.	.	40
	41	.	.	.	80
[Truncated_Name:1] 1AKE_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:2] 8BQF_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQE---				
[Truncated_Name:3] 4X8M_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:4] 6S36_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:5] 9R6U_A.pdb	VKSGSELGAQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:6] 9R71_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:7] 8Q2B_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:8] 8RJ9_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:9] 6RZE_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:10] 4X8H_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:11] 3HPR_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:12] 1E4V_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:13] 5EJE_A.pdb	VKSGSELGKQAKDIMACKLVDELVIALVKERIAQEDCR				

[Truncated_Name:14] 1E4Y_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR	
[Truncated_Name:15] 3X2S_A.pdb	VKSGSELGKQAKDIMDCGKLVDELVIALVKERIAQEDSR	
[Truncated_Name:16] 6HAP_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVRERICQEDSR	
[Truncated_Name:17] 6HAM_A.pdb	IKSGSELGKQAKDIMDAGKLVDEIIIALVKERICQEDSR	
[Truncated_Name:18] 8PVW_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR	
[Truncated_Name:19] 4K46_A.pdb	IKAGTELGKQAKSVIDAGQLVSDDIILGLVKERIAQDDCA	
[Truncated_Name:20] 4NP6_A.pdb	IKAGTELGKQAKAVIDAGQLVSDDIILGLIKERIAQADCE	
	^* *^*** *** ^~* **~*~^~^*~^*** *	
	41 . . . 80	
	81 . . . 120	
[Truncated_Name:1] 1AKE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:2] 8BQF_A.pdb	-GFLLDGFPR TIPQADAMKEAGINVDYVIEFDVPDELIVD	
[Truncated_Name:3] 4X8M_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:4] 6S36_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:5] 9R6U_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:6] 9R71_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDALIVD	
[Truncated_Name:7] 8Q2B_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:8] 8RJ9_A.pdb	NGFLLAGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:9] 6RZE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:10] 4X8H_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:11] 3HPR_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:12] 1E4V_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:13] 5EJE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:14] 1E4Y_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:15] 3X2S_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:16] 6HAP_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:17] 6HAM_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:18] 8PVW_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:19] 4K46_A.pdb	KGFLLDGFPR TIPQADGLKEVGVVVDYVIEFDVADSVIVE	
[Truncated_Name:20] 4NP6_A.pdb	KGFLLDGFPR TIPQADGLKEMGINVDYVIEFDVADDVIVE	
	**** *****^~** *^ *****^***** * ^**^	
	81 . . . 120	
	121 . . . 160	
[Truncated_Name:1] 1AKE_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:2] 8BQF_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:3] 4X8M_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:4] 6S36_A.pdb	KIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:5] 9R6U_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:6] 9R71_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:7] 8Q2B_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKA	
[Truncated_Name:8] 8RJ9_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	

[Truncated_Name:9] 6RZE_A.pdb	AIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:10] 4X8H_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:11] 3HPR_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDGTGEELTTRKD
[Truncated_Name:12] 1E4V_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:13] 5EJE_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:14] 1E4Y_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:15] 3X2S_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:16] 6HAP_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:17] 6HAM_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:18] 8PVW_A.pdb	RILKR--GETSGRV-----D
[Truncated_Name:19] 4K46_A.pdb	RMAGRRRAHLASGRTYHNVNPPKVEGKDDVTGEDLVIRED
[Truncated_Name:20] 4NP6_A.pdb	RMAGRRRAHLPSGRTYHVNVNPPKVEGKDDVTGEDLVIRED
	^ * ***
	121 . . . 160
	161 . . . 200
[Truncated_Name:1] 1AKE_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:2] 8BQF_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:3] 4X8M_A.pdb	DQEETVRKRLVEWHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:4] 6S36_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:5] 9R6U_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:6] 9R71_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:7] 8Q2B_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:8] 8RJ9_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:9] 6RZE_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:10] 4X8H_A.pdb	DQEETVRKRLVEYHQMTAALIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:11] 3HPR_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:12] 1E4V_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:13] 5EJE_A.pdb	DQEECVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:14] 1E4Y_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:15] 3X2S_A.pdb	DQEETVRKRLCEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:16] 6HAP_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:17] 6HAM_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:18] 8PVW_A.pdb	DNEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:19] 4K46_A.pdb	DKEETVLARLGVYHNQTAPLIAYYGKEAEAGNTQYLKFDG
[Truncated_Name:20] 4NP6_A.pdb	DKEETVRARLNVYHTQTAPLIEYYGKEAAAGKTQYLKFDG
	* * * * * ^ * * * * * * * * * *
	161 . . . 200
	201 . 216
[Truncated_Name:1] 1AKE_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:2] 8BQF_A.pdb	TKPVAEVRADLEKIL-
[Truncated_Name:3] 4X8M_A.pdb	TKPVAEVRADLEKILG

```

[Truncated_Name:4] 6S36_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:5] 9R6U_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:6] 9R71_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:7] 8Q2B_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:8] 8RJ9_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:9] 6RZE_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:10] 4X8H_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:11] 3HPR_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:12] 1E4V_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:13] 5EJE_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:14] 1E4Y_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:15] 3X2S_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:16] 6HAP_A.pdb     TKPVCEVRADLEKILG
[Truncated_Name:17] 6HAM_A.pdb     TKPVCEVRADLEKILG
[Truncated_Name:18] 8PVW_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:19] 4K46_A.pdb     TKAVAEVSAELEKALA
[Truncated_Name:20] 4NP6_A.pdb     TKQVSEVSADIAKALA
                                ** * ** *~* *
                                201      .      216

```

Call:

```
pdbaln(files = files, fit = TRUE, exefile = "msa")
```

Class:

```
pdbs, fasta
```

Alignment dimensions:

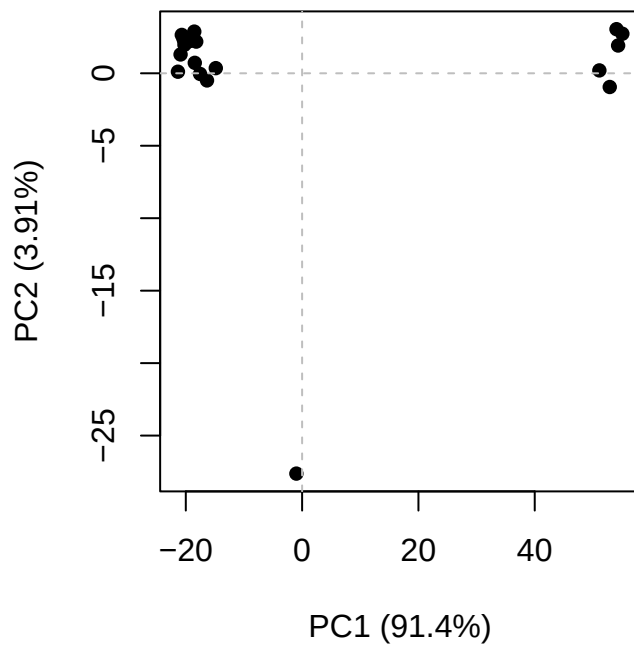
```
20 sequence rows; 216 position columns (182 non-gap, 34 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

```
#view.pdbs(pdbs)
```

PCA of all these structural data (x, y and z atom coordinates):

```
pc <- pca(pdbs)
plot(pc, 1:2)
```



Interactive view of the PC1 captured structure

```
#view.pca(pc)
```

```
mktrj(pc, file="pca.pdb")
```